Lecture #14: The Normal Approximation of the Binomial

STAT 200 – Introductory Statistics

Torin Quinlivan

Knox College

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The Meaning of Approximating Distributions

What does it mean for one distribution to "approximate" a different distribution?
The cumulative probabilities are close. That is, if X_1 and X_2 are from distributions that are "approximately" the same, then for all values of x:

$$\mathbb{P}(X_1 \leq x) \approx \mathbb{P}(X_2 \leq x)$$

Of course, there is a lot of "statistical" detail hidden in the "$\approx$" sign. How close is "close enough"? That is a question of precision. It is left up to the scientist for what precision is needed. Different applications have different needs.

Approximating the Binomial with the Normal

Let $X_1 \sim \text{Binomial}(n, p)$.
Let $X_2 \sim \text{Normal}(\mu, \sigma^2)$.
To achieve our goals, we would like to determine some rules on n, p, μ and σ^2 to ensure that

$$\mathbb{P}(X_1 \leq x) \approx \mathbb{P}(X_2 \leq x)$$

The Binomial Distribution

If $X_1 \sim \text{Binomial}(n, p)$, then:

1. The number of trials, n, is known.
2. Each trial has two possible outcomes, "success" and "not success".
3. The success probability, p, does not change from trial to trial.
4. The trials are independent.
5. The random variable is the number of successes in those n trials.

Approximating the Binomial with the Normal

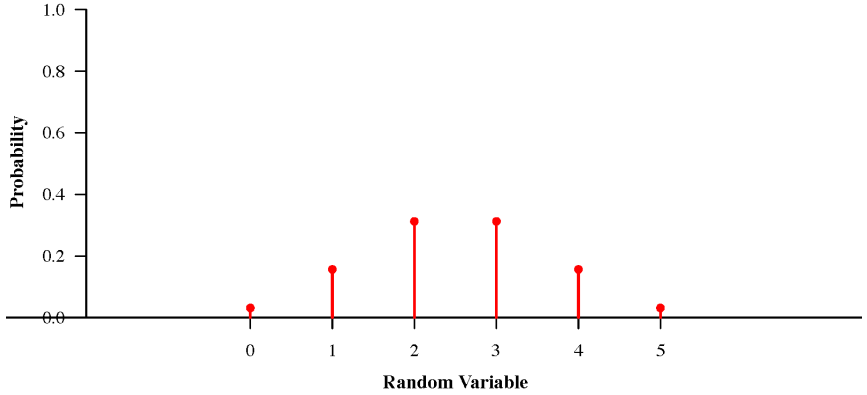
The definition of the Binomial tells us:

- $\mathbb{E}(X_1) = np$
- $\mathbb{V}(X_1) = np(1-p)$

From this, it is natural to use these parameters to define the approximate Normal distribution.
Let us see how well $X_2 \sim \text{Normal}(\mu = np, \sigma^2 = np(1-p))$ works as an approximation.

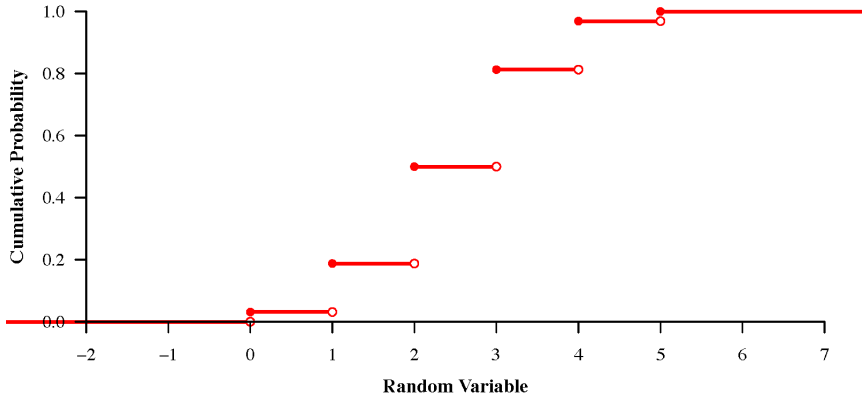
The Binomial PMF

Here is a graphic of the probability mass function (pmf) of the Binomial$(5, 0.500)$ in red. The pmf provides probabilities for a discrete distribution.



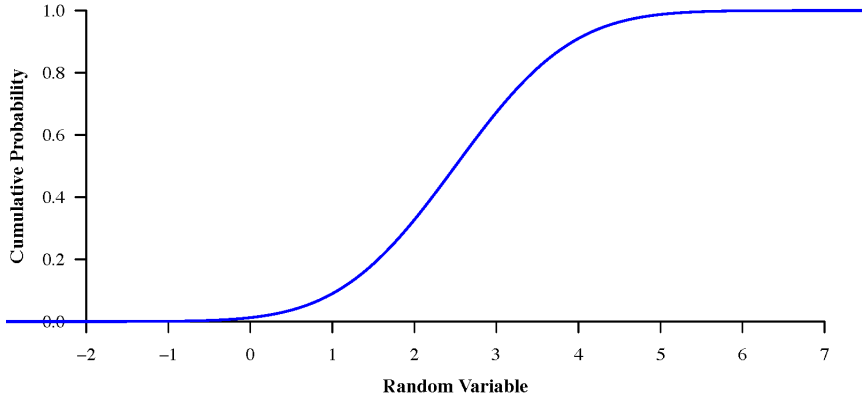
The Binomial CDF

Here is a graphic of the cumulative distribution function (CDF) of the Binomial$(5, 0.500)$ in red.



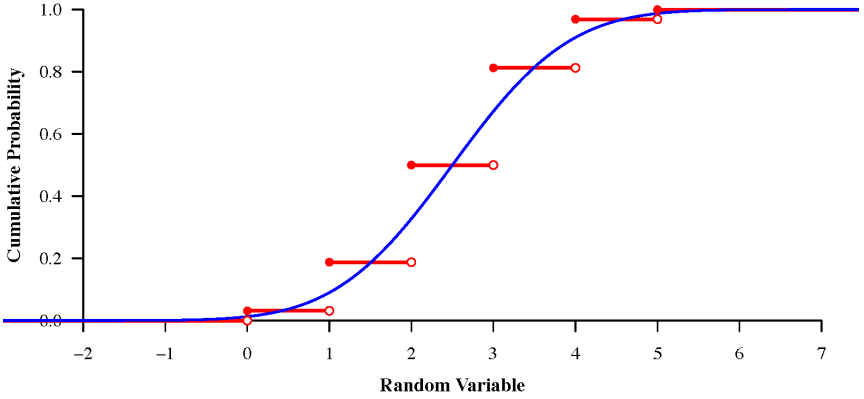
The Normal CDF

Here is a graphic of the cumulative distribution function (CDF) of the Normal($\mu = 2.5, \sigma^2 = 1.25$) in blue.



Comparing the Binomial and Normal CDFs

Here is a graphic of the cumulative distribution function (CDF) of the Binomial$(5, 0.500)$ in red and of the Normal$(\mu = 2.5, \sigma^2 = 1.25)$ in blue.



Illustrating the Effect of Sample Size

What is the effect on increasing the sample size, n, while keeping $p = 0.500$ fixed?

Several Results and Conclusions

From this brief experiment, we can come to one conclusion: The approximation is better for larger values of n.
With additional exploration, we could come to a second conclusion: The approximation is better for values of p close to 0.500.
Combining these two observations leads to the following "rule of thumb": The Normal distribution is "sufficiently" close to the Binomial distribution if both the expected number of successes and the expected number of failures are at least 10: $np \geq 10$ and $n(1-p) \geq 10$.
Note: The approximation is improved by using a "continuity correction" of 0.500. (added if $\leq$, subtracted if $\geq$, both if $=$).

Example #1: Cheating

Let us examine a certain Mathematics course taught by a certain professor at a certain midwestern college in the United States. On average, 2% of the students cheat on their examinations, thus making their transcripts all but useless as a means of illustrating their knowledge.

Use the continuity correction factor to estimate the probability that fewer than two students cheat on the midterm in that mathematics course with 50 students.

In other words, if X is the number of students cheating in the Math midterm examination, and $X \sim \text{Binomial}(50, 0.02)$, then calculate $\mathbb{P}(X < 2)$.

Example #1: Cheating

First, what is the Normal distribution that best approximates this Binomial?

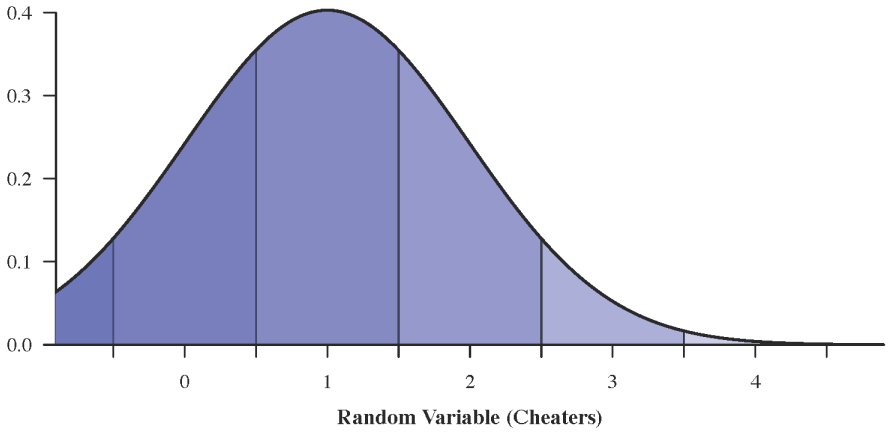
- $X \sim \text{Binomial}(50, 0.02)$.
- Thus, $\mu = np = 1$ and $\sigma^2 = np(1-p) = 0.98$.
- Then, $X \approx Y$, where $Y \sim \text{Normal}(\mu = 1, \sigma^2 = 0.98)$.

Example #1: Cheating

Next, we determine the probability statement.

- The problem is asking for $\mathbb{P}(X < 2)$.
- The continuity correction adds and/or subtracts 0.5000 to the value, **whichever includes the region**.
- Thus, $\mathbb{P}(X < 2) \approx \mathbb{P}(Y \leq 1.5)$.

Example #1: Cheating



Example #1: Cheating

Therefore, using the continuity correction, the probability of fewer than two students cheating on their Mathematics test is

```
pnorm(1.5, mean = 1, sd = sqrt(0.98))
```

```
## [1] 0.6932474
```

We can calculate the correct probability using the computer so we can compare how well the approximation did:

```
pbinom(1, size = 50, prob = 0.02)
```

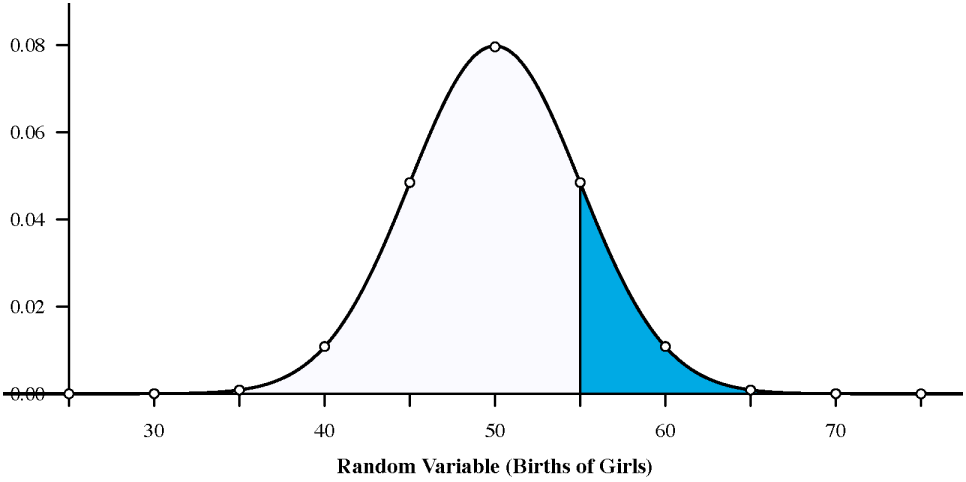
```
## [1] 0.7357714
```

Note: Here $np = 1 < 10$, so the rule of thumb is *not* met, and the approximation (0.693) is noticeably off from the exact value (0.736).

Example #2: Births

While it is not entirely accurate, for the purposes of this problem, let us assume that the probability of a girl being born is 50%. Estimate the probability of more than 55 girls being born in 100 births using a continuity correction.
In other words, if X is the number of girls born in those 100 births, and $X \sim \text{Binomial}(100, 0.500)$, then calculate $\mathbb{P}(X > 55)$.

Example #2: Births



Example #2: Births

Here are the probability calculations:

$$\begin{aligned} X &\sim \text{Binomial}(100, 0.500) \\ \implies Y &\sim \text{Normal}(\mu = 50, \sigma^2 = 25) \end{aligned}$$

$$\begin{aligned} \mathbb{P}(X > 55) &\approx \mathbb{P}(Y \geq 55.5) \\ &= 1 - \mathbb{P}(Y \leq 55.5) \\ &= 0.1356661 \end{aligned}$$

Thus, there is about a 13.6% probability that more than 55 girls will be born in 100 births.

Example #2: Births

In R, this is:

```
1 - pnorm(55.5, mean = 50, sd = 5)
```

```
## [1] 0.1356661
```

```
1 - pbinom(55, size = 100, prob = 0.500)
```

```
## [1] 0.1356265
```

Errors: The absolute error is just 0.00004, and the relative error is only 0.029%.

Example #3: The Stats Examination

After many hours of studying for your statistics examination, you believe that you have a 90% probability of answering any given question correctly. Your test includes 50 true/false questions. Assuming that your estimate is the true probability that you will answer any question correctly, use a Normal distribution to estimate the probability that you will miss no more than 4 questions.

In other words, if X is the number of problems you get **wrong** on this examination, and $X \sim \text{Binomial}(50, 0.100)$, then calculate $\mathbb{P}(X \leq 4)$.

Example #3: The Stats Examination

Here are the probability calculations:

$$\begin{aligned} X &\sim \text{Binomial}(50, 0.100) \\ \implies Y &\sim \text{Normal}(\mu = 5, \sigma^2 = 4.5) \\ \mathbb{P}(X \leq 4) &\approx \mathbb{P}(Y \leq 4.5) \\ &= 0.4068319 \end{aligned}$$

Thus, the probability that you miss no more than four problems, or that you will earn at least an A- is 40.7%. What is the probability you'll earn an A?

Example #3: The Stats Examination

In R, this is:

```
pnorm(4.5, mean = 5, sd = sqrt(4.5))
```

```
## [1] 0.4068319
```

```
pbinom(4, size = 50, prob = 0.100)
```

```
## [1] 0.4311984
```

Errors: The absolute error is -0.024367, and the relative error is -5.650%. This is much worse than the other examples, because $np = 5 < 10$: the rule of thumb is not met.

Example #4: Toothpaste

Many toothpaste commercials advertise that "3 out of 4 dentists recommend their brand of toothpaste." Use a Normal distribution to estimate the probability that in a random survey of 400 dentists, exactly 300 will recommend Brand F toothpaste.
In this calculation, let us assume that the commercials are correct, and therefore, there is a 75% chance that any specific dentist will recommend Brand F toothpaste.
In other words, if X is the number of dentists recommending Brand F toothpaste, and $X \sim \text{Binomial}(400, 0.750)$, then calculate $\mathbb{P}(X = 300)$.

Example #4: Toothpaste

Here are the probability calculations:

$$\begin{aligned} X &\sim \text{Binomial}(400, 0.750) \\ \implies Y &\sim \text{Normal}(\mu = 300, \sigma^2 = 75) \end{aligned}$$

$$\begin{aligned} \mathbb{P}(X = 300) &\approx \mathbb{P}(299.5 < Y \leq 300.5) \\ &\approx \mathbb{P}(Y \leq 300.5) - \mathbb{P}(Y \leq 299.5) \\ &\approx 0.0460403 \end{aligned}$$

Thus, the probability that *exactly* 300 randomly-selected dentists recommend Brand F toothpaste is approximately 0.0460403.

Example #4: Toothpaste

In R, this is:

```
pnorm(300.5, mean = 300, sd = sqrt(75)) - pnorm(299.5, mean = 300, sd = sqrt(75))
```

```
## [1] 0.04604031
```

```
dbinom(300, size = 400, prob = 0.75)
```

```
## [1] 0.04602432
```

Errors: The absolute error is just 0.000016, and the relative error is only 0.034%.